

Assignment

①

Optimization for Data Science

Facial 2

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Q1) Consider problem

$$\min 2x_1 - x_2 + 2x_3$$

s.t

$$2x_1 + x_2 + x_3 = 4$$

$$x_1 + 2x_2 + x_4 = 5$$

$$x_1, x_2, x_3, x_4 \geq 0$$

Solution

We can follow algebraic approach by getting $n-m = 4-2 = 2$ non-basic variables to zero & solve the remaining var.

Case 1: put $x_3 = 0, x_4 = 0$, constraint change

$$2x_1 + x_2 = 4 \quad - (I)$$

$$x_1 + 2x_2 = 5 \quad - (II)$$

multiply eqⁿ (II) by 2 & subtract eqⁿ (I)

(2)

$$\begin{array}{r} 2x_1 + 4x_2 = 10 \\ -2x_1 - x_2 = -4 \\ \hline \end{array}$$

$$3x_2 = 6 \Rightarrow x_2 = 2$$

put x_2 in eq. (1)

$$x_1 + 2(2) = 5 \Rightarrow x_1 = 1$$

Solution (1, 2, 0, 0)

Case 2: put $x_2 = 0, x_3 = 0$

$$2x_1 = 4 \Rightarrow x_1 = 2$$

$$x_1 + x_4 = 5 \Rightarrow x_4 = 5 - 2 = 3$$

Solution (2, 0, 0, 3)

Case 3: $x_2 = 0, x_4 = 0$

$$2x_1 + x_3 = 4 \quad - (1)$$

$$x_1 = 5 \quad - (11)$$

put $x_1 = 5$ in (1)

$$2(5) + x_3 = 4 \Rightarrow x_3 = -6$$

Not feasible as $x_3 < 0$

Case 4: $x_1 = 0, x_3 = 0$

$$x_2 = 4$$

$$2x_2 + x_4 = 5 \Rightarrow 2(4) + x_4 = 5$$

$$x_4 = -3$$

(3)

Not feasible sol as $x_4 < 0$

Case 5: $x_1 = 0, x_4 = 0$

$$x_2 + x_3 = 4 \quad \text{--- (1)}$$

$$2x_2 = 5 \Rightarrow x_2 = 2.5$$

put x_2 in (1)

$$2.5 + x_3 = 4 \Rightarrow x_3 = 1.5$$

solution $(0, 2.5, 1.5, 0)$

Case 6: $x_1 = 0, x_2 = 0$

$$x_3 = 4$$

$$x_4 = 5$$

solution $(0, 0, 4, 5)$

So all feasible solution are

$$(1, 2, 0, 0)$$

$$(2, 0, 0, 3)$$

$$(0, 2.5, 1.5, 0)$$

$$(0, 0, 4, 5)$$

optimal solution, put all these value in

$$Z = 2x_1 - x_2 + 2x_3$$

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$$(1, 2, 0, 0)$$

$$Z = 2(1) - 2(2) + 2(0) = 2 - 2 = 0$$

$$(2, 0, 0, 3)$$

$$Z = 2(2) - 0 + 2(0) = 4$$

$$(0, 2.5, 1.5, 0)$$

$$Z = 2(0) - 2.5 + 2(1.5) = -2.5 + 3 = 0.5$$

$$(0, 0, 4, 5)$$

$$Z = 2(0) - 0 + 2(4) = 8$$

$$Z(\min) = 0 \quad \text{at } (1, 2, 0, 0)$$

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$$(1, 2, 0, 0)$$

is optimal solution

Q2)

①

$$\max Z = 3x_1 + 2x_2$$

Solution

change the problem to equality
adding slack variable in constraint

$$2x_1 + x_2 + S_1 = 6$$

$$x_1 + 2x_2 + S_2 = 6$$

create simplex table from problem
with objective function

$$Z = 3x_1 + 2x_2 + 0S_1 + 0S_2$$

Row	CB	C(j)	3	2	0	0	Solution	Ratio
		BV	x_1	x_2	S_1	S_2		
R1	0	S_1	2	1	1	0	6	3
R2	0	S_2	1	2	0	1	6	6
		Z(j)	0	0	0	0	0	
		Z(j) - C(j)	3	2	0	0		

← (min)

(max) ↑

Condⁿ $Z(j) - C(j) \leq 0$ is failed

max of $Z_j - C_j$ is pivot column & Ratio
with least positive is pivot row

⑥

$$\text{pivot element} = 2$$

$$\text{entering var} = x_1$$

$$\text{exit var} = S_1$$

operation to make pivot ele 1 & rest column element zero

$$R_1 \rightarrow R_1 / \text{Pivot} = R_1 / 2$$

$$R_2 \rightarrow R_2 - \text{New } R_1 \times 1$$

Now Simplex table is

Row	CB	Cj	3	2	0	0	Solution	Ratio
		BV	x_1	x_2	S_1	S_2		
R1	3	x_1	1	1/2	1/2	0	3	6
R2	0	S_1	0	3/2	-1/2	1	3	2 ←
		Z(j)	3	3/2	3/2	0	9	
		$C_j - Z_j$	0	1/2	3/2	0		

↑

$$\text{pivot ele} = 3/2$$

$$\text{entering} \Rightarrow x_2$$

$$\text{exit} \Rightarrow S_2$$

operation

$$R_2 \rightarrow R_2 \times 2/3$$

$$R_1 \rightarrow R_1 - \text{New } R_2 / 2$$

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Row	CB	C _j	3	2	0	0	Solution	Ratio
		BV	x ₁	x ₂	s ₁	s ₂		
R1	3	x ₁	1	0	2/3	-1/3	2	
R2	2	x ₂	0	1	-1/3	2/3	2	
		Z _j	3	2	1/3	1/3	10	
		C _j - Z _j	0	0	-1/3	-1/3		

$$C_j - Z_j \leq 0 \quad (\text{pass})$$

$$\Rightarrow x_1 = 2, x_2 = 2, Z(\max) = 10$$

put these value in function 2

$$Z = 3 \times 2 + 2 \times 2 = 6 + 4 = 10$$

(verify).

Hence proved

(2, 2) is optimal solution

with $Z(\max) = 10$.

(i)

$$\begin{aligned} \max \quad & x_1 + 2x_2 + 2x_3 \\ \text{s.t.} \quad & 2x_1 + x_2 + x_3 \leq 6 \\ & x_2 + x_3 \leq 4 \end{aligned}$$

Solution

change the problem to equality, adding slack variable in constraint

$$\begin{aligned} 2x_1 + x_2 + x_3 + S_1 &= 6 \\ x_2 + x_3 + S_2 &= 4 \end{aligned}$$

$$Z = x_1 + 2x_2 + 2x_3 + 0S_1 + 0S_2$$

Create simplex table

Row	CB	C _j	1	2	2	0	0	Solution	Ratio
		BV	x ₁	x ₂	x ₃	S ₁	S ₂		
R ₁	0	S ₁	2	1	1	1	0	6	6
R ₂	0	S ₂	0	1	1	0	1	4	4
		Z _j	0	0	0	0	0	0	
	C _j - Z _j		1	2	2	0	0		

↑

pivot element $\Rightarrow$ 1entering $\Rightarrow$ x₂exit $\Rightarrow$ S₂

operation

$$R_2 = R_2 / 1 = R_2$$

$$R_1 = R_1 - \text{New } R_2 \times 1$$

So table will be

Row & CB	C _j	BV	1	2	2	0	0	Solution	Ratio
			x ₁	x ₂	x ₃	s ₁	s ₂		
R ₁	0	s ₁	2	0	0	1	-1	2	1 ←
R ₂	2	x ₂	0	1	1	0	1	4	—
		Z _j	0	2	2	0	2	8	
		C _j - Z _j	1	0	0	0	-2		

Condⁿ $C_j - Z_j \leq 0$ (Failed)

So pivot var $\Rightarrow 2$

entering variable $\Rightarrow x_1$

exit variable $\Rightarrow s_1$

operation

$$R_1 \rightarrow R_1 / 2$$

$$R_2 \rightarrow R_2$$

So simplex table become

Row	CB	C_j	1	2	2	0	0	Solution	Ratio
		BV	x_1	x_2	x_3	s_1	s_2		
R_1	1	x_1	1	0	0			1	
R_2	2	x_2	0	1	1			4	
		Z_j	1	2	2			9	
		$C_j - Z_j$	0	0	0				

$$C_j - Z_j \leq 0 \quad \text{pass}$$

so we will stop here

$$x_1 = 1, x_2 = 4, x_3 = 0 \quad \& \quad Z = 9$$

put x_1, x_2 & x_3 in main objective fun.

$$x_1 + 2x_2 + 2x_3$$

$$1 + 2 \times 4 + 2 \times 0 = 9$$

hence proved

(1, 4, 0) is optimal solution

$$\& \quad \mathbf{Z(\max) = 9}$$

(11)

Q3)

$$\min f(x, y) = x^2 + y^2$$

$$\text{s.t. } x + y = 1$$

Solution

$$f(x, y) = x^2 + y^2$$

$$g(x, y) = x + y - 1$$

Lagrange

$$L = f(x, y, \lambda) = f(x, y) + \lambda(g, x)$$

To solve the above eqⁿ we do partial Derivatives

$$L = x^2 + y^2 + \lambda(x + y - 1) \quad - (i)$$

$$\frac{dL}{dx} = 2x + \lambda = 0 \Rightarrow \lambda = -2x \quad - (ii)$$

$$\frac{dL}{dy} = 2y + \lambda = 0 \Rightarrow \lambda = -2y \quad - (iii)$$

$$\frac{dL}{d\lambda} = x + y - 1 = 0 \quad - (iv)$$

$\Rightarrow$ from (ii) & (iii)

$$-2x = -2y \Rightarrow x = y \quad - (v)$$

put in (iv)

$$x + x - 1 \Rightarrow x = 1/2$$

$$\Rightarrow \text{from (v)} \quad y = 1/2$$

$$\Rightarrow \text{from (ii)} \quad \lambda = -1$$

So critical points are $(1/2, 1/2)$

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Now, calculating the Border Hessian Matrix

$$P(\lambda) = \begin{vmatrix} \Phi - \lambda I & H \\ H^T & 0 \end{vmatrix} = 0$$

$$\Phi = \begin{bmatrix} d^2L/dx^2 & d^2L/dxdy \\ d^2L/dxdy & d^2L/dy^2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$H = \begin{bmatrix} dg/dx \\ dg/dy \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\Phi - \lambda I = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2-\lambda & 0 \\ 0 & 2-\lambda \end{bmatrix}$$

$$P(\lambda) = \begin{bmatrix} 2-\lambda & 0 & 1 \\ 0 & 2-\lambda & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\text{Determinant} = (2-\lambda)((2-\lambda)(0) - 1(1)) - 0 + 1(0 \times 1 - (2-\lambda)(1))$$

$$= (2-\lambda)(-1) + (-2+\lambda)$$

$$= -2+\lambda -2+\lambda = -4+2\lambda = 0$$

$$2\lambda = 4$$

$$\lambda = 2 (+ve)$$

Here λ is +ve
So it's (minima)

$(1/2, 1/2)$ is optimal solution

$$z(\max) = x^2 + y^2$$

$$\text{put } x = 1/2, y = 1/2$$

$$z = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{4} + \frac{1}{4}$$

$$z(\max) = \frac{1}{2}$$

$$\text{at } (1/2, 1/2)$$

Q4)

solve the problem using a penalty function method

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$$\min x^2 + (y-2)^2$$

$$\text{s.t. } y = x+1$$

Solution

$$f(x) = z = x^2 + (y-2)^2$$

$$g(x) \Rightarrow y = x+1$$

$$y - x - 1 = 0$$

by Penalty function

$$P(x, y, r) = f(x, y) + r g(x, y)^2$$

put values

$$P = x^2 + (y-2)^2 + r(y-x-1)^2 \quad \text{--- (1)}$$

Now do partial diff.

$$\frac{dP}{dx} = 2x - 2r(y-x-1) = 0 \quad \text{--- (2)}$$

$$\frac{dP}{dy} = 2(y-2) + 2r(y-x-1) = 0 \quad \text{--- (3)}$$

Solve eqⁿ (2) & (3)

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Solve (2)

$$2x - 2y(y - x - 1) = 0$$

$$2x - 2xy + 2yx + 2y = 0$$

$$2x(1+y) - 2y(y+1) = 0$$

$$x = \frac{(y-1)y}{1+y} \quad - (4)$$

Solve (3)

$$2(y-2) + 2x(y-x-1) = 0$$

$$2y - 4 + 2xy - 2xx - 2x = 0$$

$$y - 2 + xy - xx - x = 0$$

put x from (4)

$$y(1+y) - 2 - x - x\left(\frac{(y-1)y}{1+y}\right) = 0$$

$$y(1+y)(1+y) - (2+y)(1+y) - x^2(y-1) = 0$$

$$y + y^2 + 2yx - 2 - 3x - x^2 - xy + x^2 = 0$$

$$y + 2yx - 2 - 3x = 0$$

$$y(1+2x) - (2+3x) = 0$$

$$y = \frac{2+3x}{1+2x} \quad - (5)$$

put y in eqn (4)

$$x = \frac{(2+3x-1-2x)x}{(1+x)(1+2x)} = \frac{(1+x)x}{(1+x)(1+2x)}$$

$$= \frac{x}{1+2x} \quad - \textcircled{6}$$

as x approaches to ∞

$$\lim_{x \rightarrow \infty} x = \lim_{x \rightarrow \infty} \frac{x}{1+2x} = \frac{\cancel{x}/\cancel{x}}{\cancel{x} \frac{1}{x} + \frac{2x}{\cancel{x}}}$$

$$= \lim_{x \rightarrow \infty} \frac{1}{2 + 1/x}$$

$$= \frac{1}{2+0} = 1/2$$

$$\lim_{x \rightarrow \infty} y = \lim_{x \rightarrow \infty} \frac{2+3x}{1+2x} = \lim_{x \rightarrow \infty} \frac{2/x + 3\cancel{x}/\cancel{x}}{1/x + 2\cancel{x}/\cancel{x}}$$

$$= \frac{0+3}{0+2} = 3/2$$

So $(1/2, 3/2)$ is solution (optimal)

$$\& \min(z) = \left(\frac{1}{2}\right)^2 + \left(\frac{3}{2} - 2\right)^2 = \frac{1}{4} + \frac{1}{4}$$

$$\min(z) = 1/2$$

An

Q5)

$$\min x^4 + y^2$$

$$\text{s.t. } x \geq 0, y \geq 0$$

Solution

$$f(x, y) = \max x^4 + y^2$$

$$B(x, y, \gamma) = f(x) + \gamma B(x)$$

$$B(x) = \sum_{i=1}^n \left(\frac{-1}{g_i(x)} \right)$$

$$B(x) = \frac{-1}{(-x)} + \frac{-1}{(-y)} = \left(\frac{1}{x} + \frac{1}{y} \right)$$

$$F(x, y, \gamma) = x^4 + y^2 + \gamma \left(\frac{1}{x} + \frac{1}{y} \right) \quad - (i)$$

$$\frac{dF}{dx} = 4x^3 + \gamma \left(\frac{-1}{x^2} \right) = 0$$

$$= 4x^5 - \gamma = 0 \Rightarrow x^5 = \gamma/4 \quad - (ii)$$

$$\frac{dF}{dy} = 2y + \gamma \left(\frac{-1}{y^2} \right) = 0$$

$$= 2y^3 - \gamma = 0 \Rightarrow y^3 = \gamma/2 \quad - (iii)$$

as barrier funcⁿ tends to zero (0)

put $\lim \gamma \rightarrow 0$

from (2)

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$$\lim_{y \rightarrow 0} x = 0/y = 0$$

from (3)

$$\lim_{x \rightarrow 0} y = 0/2 = 0$$

So $(0,0)$ is optimal solution for
method $x^4 + y^2$
with min objective function

$$z = 0^4 + 0^2 = 0$$

Solution $(0,0)$ & min value 0.

Q6)

Frank & Wolfe

$$\min x^3 + y^3$$

$$\text{s.t. } \begin{aligned} 3x + y &\leq 3 \\ 5x - 3y &\leq 5, \quad x, y \geq 0 \end{aligned}$$

Solution

Iteration 1

① initial feasible solution at

$$(x_0, y_0) = (0, 0)$$

② $\nabla f(x, y) = \left(\frac{df}{dx}, \frac{df}{dy} \right)$ Linearize the obj func

$$\nabla f(x, y) = (3x^2, 3y^2)$$

$$\nabla f(x_0, y_0) = (0, 0)$$

$$\therefore f(x) = 0 \cdot x + 0 \cdot y = 0$$

the gradient is zero at any feasible point will yield same value

Now, consider $(x_0, y_0) = (0, 1)$

$$\nabla f(x_0, y_0) = (3 \cdot (0)^2, 3 \cdot (1)^2)$$

$$\nabla(0, 1) = (0, 3)$$

feasible $3x + y \leq 3$

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$$3(0) + 1 \leq 3$$

$$1 \leq 3$$

$$5x - 3y \leq 5$$

$$5(0) - 3(1) \leq 5$$

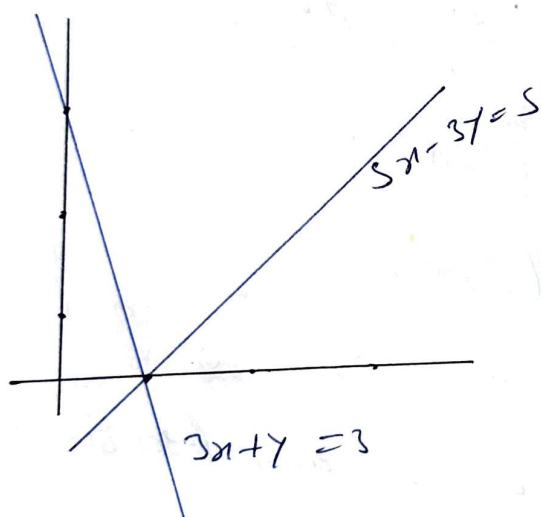
$$-3 \leq 5$$

Linear objective $0.2x + 3y = 3y$

Now minimize $3y$

$$\text{s.t. } 3x + y \leq 3 \quad x, y \geq 0$$

$$5x - 3y \leq 5$$



$$\begin{array}{l} \underline{3y} \\ (0, 0) : 3(0) = 0 \\ (0, 3) = 3(3) = 9 \\ (1, 0) = 3(0) = 0 \end{array}$$

check $(1, 0)$

$$3x + y = 3 \Rightarrow y = 3 - 3x$$

$$5x - 3y = 5 \quad \text{put } (y)$$

$$5x - 3(3 - 3x) = 5 \Rightarrow 5x - 9 + 9x = 5$$

$$14x = 14 \Rightarrow x = 1$$

$$y = 3 - 3(1) = 0$$

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$$x(\alpha), y(\alpha) = (0,1) + \alpha((1,0) - (0,1)) = (\alpha, 1-\alpha)$$

$$g(\alpha) = f(x(\alpha), y(\alpha)) = \alpha^3 + (1-\alpha)^2$$

min $g(\alpha)$ find $g'(\alpha)$

$$g'(\alpha) = 3\alpha^2 - 2(1-\alpha) = 0$$

$$\Rightarrow 3\alpha^2 - 2(1-\alpha) = 0$$

$$3\alpha^2 - 2 + 2\alpha = 0 \Rightarrow \alpha = 1/2$$

$$0 \leq \alpha \leq 1$$

$$(x_1, y_1) = (0,1) + \frac{1}{2}((1,0) - (0,1)) = (0,1) + \left(\frac{1}{2}, -\frac{1}{2}\right) \\ = \left(\frac{1}{2}, \frac{1}{2}\right)$$

Iteration 2 $x_1 = 1/2, y_1 = 1/2$

Unique obj f^*

$$\nabla f(1/2, 1/2) = 3\left(\frac{1}{4}\right), 3\left(\frac{1}{4}\right) \\ = \left(\frac{3}{4}, \frac{3}{4}\right)$$

$$Z = \frac{3}{4}x + \frac{3}{4}y$$

$$\text{min } Z \quad \text{s.t.} \quad 3x + y \leq 3 \quad x, y \geq 0 \\ 5x - 3y \leq 5$$

$$\text{At } (0,0) = 0$$

$$(1,0) = 1 + 0 = 1$$

$$(0,3) = 0 + 3 = 3$$

$$(1,0) = 1+0 = 1$$

(20)

min occurs at $(0,0)$

$$\begin{aligned} x(\alpha), y(\alpha) &= \frac{1}{2}, \frac{1}{2} + \alpha((0,0) - (\frac{1}{2}, \frac{1}{2})) \\ &= (\frac{1}{2} - \frac{\alpha}{2}, \frac{1}{2} - \frac{\alpha}{2}) \end{aligned}$$

$$\begin{aligned} g(\alpha) &= f(x(\alpha), y(\alpha)) \\ &= (\frac{1}{2} - \frac{\alpha}{2})^3 + (\frac{1}{2} - \frac{\alpha}{2})^3 \\ &= 2(\frac{1}{2} - \frac{\alpha}{2})^3 \end{aligned}$$

min $g(\alpha)$ & cal $g'(\alpha)$

$$g'(\alpha) = 6(\frac{1}{2} - \frac{\alpha}{2})^2(-\frac{1}{2}) = 0$$

$$\alpha = 1$$

condⁿ is true only in the condⁿ $0 \leq \alpha \leq 1$

So final solution is

$$\begin{aligned} f(x_c, y_c) &= (\frac{1}{2}, \frac{1}{2}) + 1((0,0) - (\frac{1}{2}, \frac{1}{2})) \\ &= (\frac{1}{2}, \frac{1}{2}) + (-\frac{1}{2}, -\frac{1}{2}) \\ &= (0,0) \end{aligned}$$

After 2 iteration $(0,0)$ is optimal solⁿ of given objective funcⁿ.

$$\min Z = 0$$

Q7)

Solve LPP using Charnik - M

$$\max Z = 8x + 4y$$

$$s.t \quad 4x + 2y \leq 30$$

$$2x + 4y \geq 24$$

$$x, y \geq 0$$

Solution

Convert constraint to equalities

$$4x + 2y + S_1 = 30$$

$$2x + 4y - S_2 + A_1 = 24$$

$$\max Z = 8x + 4y + 0 \cdot S_1 + 0 \cdot S_2 + M \cdot A_1$$

Initial table

Row	C _j		8	4	0	0	-M	Solution	Ratio
	CB	BV	x ₁	x ₂	S ₁	S ₂	A ₁		
R ₁	0	S ₁	4	2	1	0	0	30	15
R ₂	-M	A ₁	2	4	0	-1	1	24	6 ←
		Z _j	-2M	-4M	0	M	M	9M	
		C _j - Z _j	8+2M	4+4M	0	-M	-M	9M	

↑

 $C_j - Z_j \leq 0$ (Feasible)

pivot variable $\Rightarrow 4$
element

entering variable x_2
exit variable A_1

Rotation

$$R_2 \rightarrow R_2 / \text{pivot} = R_2 / 4$$

$$R_1 \rightarrow R_1 - \text{New } R_2 \times 2$$

Iteration 1

	C_j		0	4	0	0		
Row & CB	OV	x_1	x_2	S_1	S_2	Solution	R_{ratio}	
R1	0	S_1	3	0	1	1/2	18	6 $\leftarrow$
R2	4	x_2	1/2	1	0	-1/4	6	12
Z_j	-		2	4	0	-1	24	
$C_j - Z_j$			6	0	0	1		

Condition $C_j - Z_j \leq 0$ (failed)

pivot element 3

entering variable x_1

exit variable S_1

Rotation

$$R_1 \rightarrow R_1 / 3$$

$$R_2 \rightarrow R_2 - \text{New } R_1 \times \frac{1}{2}$$

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Iteration 2

Row #	CB	C_j	θ	4	0	Solution	Ratio
		BV	x_1	x_2	S_2		
Row 1	8	x_1	1	0	3/2	6	
Row 2	4	x_2	0	1	0	3	
Z_j			8	4	12	60	
$C_j - Z_j$			0	0	-12		

$$C_j - Z_j \leq 0 \quad (\text{Pass})$$

So we will stop here.

optimal points are (6, 3) with $Z_{\max} = 60$

put $x_1 = 6, x_2 = 3$ in function

$$\Rightarrow 8x_1 + 4x_2$$

$$\Rightarrow 8 \times 6 + 4 \times 3 = 48 + 12$$

$$= 60$$

verified

(6, 3) is optimal solution

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$$\min x_1^2 + x_2^2 + 2x_3 \quad - (1)$$

S.t

$$x_1 + x_2 + x_3 = 6 \quad - (11)$$

$$-x_1 + x_2 + x_3 = 4 \quad - (3)$$

Solution 8

$$L(x_1, x_2, x_3, \lambda) = x_1^2 + x_2^2 + 2x_3 + \lambda_1(x_1 + x_2 + x_3 - 6) + \lambda_2(-x_1 + x_2 + x_3 - 4)$$

$$\frac{dL(x_1)}{dx_1} = 2x_1 + \lambda_1 - \lambda_2 \quad - (4) \quad 2x_1 = \lambda_2 - \lambda_1$$

$$\frac{dL}{dx_2} = 2x_2 + \lambda_1 + \lambda_2 \quad - (5) \quad -2x_2 = -\lambda_2 + \lambda_1$$

$$\frac{dL}{dx_3} = 2 + \lambda_1 + \lambda_2 \quad - (6) \quad \lambda_1 + \lambda_2 = -2$$

$$\frac{dL}{d\lambda_1} = x_1 + x_2 + x_3 - 6 \quad - (7)$$

$$\frac{dL}{d\lambda_2} = -x_1 + x_2 + x_3 - 4 \quad - (8)$$

$$\text{put } (4), (5), (6), (7), (8) = 0$$

$$\text{put } \lambda_1 + \lambda_2 = -2 \text{ from } (6) \text{ in eqn } (5)$$

$$-2x_2 = -\lambda_2 \Rightarrow x_2 = 1$$

put x_2 in ⑦ & ⑧

$$\begin{aligned} x_1 + 1 + x_3 - 6 &= 0 \Rightarrow x_1 + x_3 = 5 \quad \text{--- ⑨} \\ -x_1 + 1 + x_3 - 4 &= 0 \Rightarrow \frac{x_1 + x_3 = 5}{-x_1 + x_3 = 3} \\ &\quad \quad \quad 2x_3 = 8 \end{aligned}$$

$$x_3 = 4$$

$$\text{from ⑨} \quad x_1 + 4 = 5 \Rightarrow x_1 = 1$$

points are $(1, 1, 4)$ using Lagrange Method.

now calculate λ_1 & λ_2

$$\begin{aligned} \lambda_2 - \lambda_1 &= 2x_1 \\ \Rightarrow \lambda_2 - \lambda_1 &= 2 \\ \lambda_1 + \lambda_2 &= -2 \\ \hline 2\lambda_2 &= 0 \Rightarrow \lambda_2 = 0 \end{aligned}$$

$$\lambda_1 = -2$$

verify KKT points

$$1) \text{ Stationary } \nabla f = \lambda_1 \nabla g_1 + \lambda_2 \nabla g_2$$

$$f(x_1, x_2, x_3) = x_1^2 + x_2^2 + 2x_3$$

$$\nabla f(x_1, x_2, x_3) = 2x_1 + 2x_2 + 2$$

$$g_1 = x_1 + x_2 + x_3 - 6, \quad g_2 = -x_1 + x_2 + x_3 - 4$$

$$\nabla g_1 = (1, 1, 1), \quad \nabla g_2 = (-1, 1, 1)$$

$$\text{check } \nabla f = \lambda_1 \nabla g_1 + \lambda_2 \nabla g_2$$

$$(2, 2, 2) = 2(1, 1, 1) + 0(-1, 1, 1) \\ = (2, 2, 2) \quad \text{verify}$$

2) primal feasibility

$$\textcircled{1} \quad x_1 + x_2 + x_3 = 6$$

$$\text{put } x_1 = 1, x_2 = 1, x_3 = 4$$

$$1 + 1 + 4 = 6$$

$$6 = 6 \quad \text{verify}$$

$$\textcircled{2} \quad -x_1 + x_2 + x_3 = 4$$

$$-1 + 1 + 4 = 4$$

$$4 = 4$$

3) Dual feasibility

No equality constrained. So always satisfy

$$\text{So } (x_1, x_2, x_3) = (1, 1, 4)$$

This points are KKT points & since all KKT condition satisfied.

$$Z(\min) = x_1^2 + x_2^2 + 2x_3$$

$$= 1^2 + 1^2 + 2 \times 4 = 2 + 8 = 10$$

So objective fn is minimum 10 at critical point $(1, 1, 4)$

solution

Q9

$$\min 3x_1^2 + x_2^2 + 2x_3^2 + 2x_1 + 2x_2 + 2x_3$$

$$\text{s.t. } x_1^2 + x_2^2 + x_3^2 = 1$$

find KKT point

Solution

$$Z = 3x_1^2 + x_2^2 + 2x_3^2 + 2x_1 + 2x_2 + 2x_3$$

$$g(x) = x_1^2 + x_2^2 + x_3^2 - 1 = 0$$

Lagrangian function

$$L(x_1, x_2, x_3, \lambda) = 3x_1^2 + x_2^2 + 2x_3^2 + 2x_1 + 2x_2 + 2x_3 + \lambda(x_1^2 + x_2^2 + x_3^2 - 1)$$

$$\frac{dL}{dx_1} = 6x_1 + 2 + 2\lambda x_1 = 0$$

$$2(3x_1 + 1 + \lambda x_1) = 0 \Rightarrow (3 + \lambda)x_1 = -1$$

$$x_1 = -1/(3 + \lambda)$$

$$\frac{dL}{dx_2} = 2x_2 + 2 + 2\lambda x_2 = 0$$

$$2(x_2 + 1 + \lambda x_2) = 0 \Rightarrow (1 + \lambda)x_2 = -1$$

$$x_2 = -1/(1 + \lambda)$$

$$\frac{dL}{dx_3} = 4x_3 + 2 + 2\lambda x_3 = 0$$

$$(4 + 2\lambda)x_3 = -2 \Rightarrow (2 + \lambda)x_3 = -1$$

$$x_3 = -1/(2 + \lambda)$$

KKT Condⁿ require

$$\lambda(x_1^2 + x_2^2 + x_3^2 - 1) = 0$$

Case 1

$$\text{pod } \lambda = 0$$

 $x_1, x_2, x_3 \rightarrow \infty$ is invalid
Case 2

$$x_1^2 + x_2^2 + x_3^2 = 1$$

$$\left(\frac{-1}{3+\lambda}\right)^2 + \left(\frac{-1}{1+\lambda}\right)^2 + \left(\frac{-1}{2+\lambda}\right)^2 = 1$$

$$(1+\lambda)(2+\lambda) + (2+\lambda)(3+\lambda) + (1+\lambda)(3+\lambda) = (1+\lambda)(2+\lambda)(3+\lambda)$$

$$\lambda \approx -4.1991, \quad \lambda \approx 0.1991$$

Case 1

$$\lambda = -4.1991$$

$$x_1 = \frac{-1}{3+\lambda} = \frac{1}{3-4.19} = 0.834$$

$$x_2 = \frac{-1}{1+\lambda} = \frac{-1}{1-4.19} = 0.313$$

$$x_3 = \frac{-1}{(2+\lambda)} = \frac{-1}{2-4.19} = 0.455$$

$$(0.834, 0.313, 0.455)$$

Case 2

$$\lambda = 0.19$$

$$x_1 = \frac{-1}{3+0.19} = -0.313$$

$$x_2 = \frac{-1}{1+0.19} = -0.834$$

$$u_3 = \frac{-1}{2+1} = \frac{-1}{2+0.19} = -0.455$$

$$(-0.313, -0.834, -0.455)$$

Verify constraint

$$u_1^2 + u_2^2 + u_3^2 = 1$$

$$(0.834)^2 + (0.313)^2 + (0.455)^2 = 1$$

$$1 \approx 1$$

$$(-0.313)^2 + (-0.834)^2 + (0.455)^2 = 1$$

$$1 \approx 1$$

Both verify KKT condⁿ Now check optimality

$$\lambda = -4.19, (u_1, u_2, u_3) = (0.83, 0.31, 0.45)$$

$$F = 3(0.83)^2 + (0.31)^2 + 2(0.45)^2 + 2(0.833) + 2(0.31) + 2(0.454)$$

$$= 2.068 + 0.0971 + 0.41 + 0.62 + 0.9 + 1.66$$

$$\approx 5.8$$

$$\lambda = -0.19, (-0.313, -0.83, -0.45)$$

$$F = 3(-0.313)^2 + (-0.83)^2 + 2(-0.45)^2 + 2(-0.313) + 2(-0.83) + 2(-0.45)$$

$$= -1.8$$

So, $\lambda = 0.19$ satisfy the constraint
equation Cond^h ,

So $\lambda = 0.19$ is KKT multiplier

Δ KKT points are

$(-0.313, -0.83, -0.45)$